Class 5: Data Viz With gglot

Shawn Xiao(PID: A18276668)

Today we are exploring the **ggplot** package and how to make nice figures in R.

There are lots of ways to make figures and plots in R. These include:

- so called "base" R
- and add on packages like **ggplot**

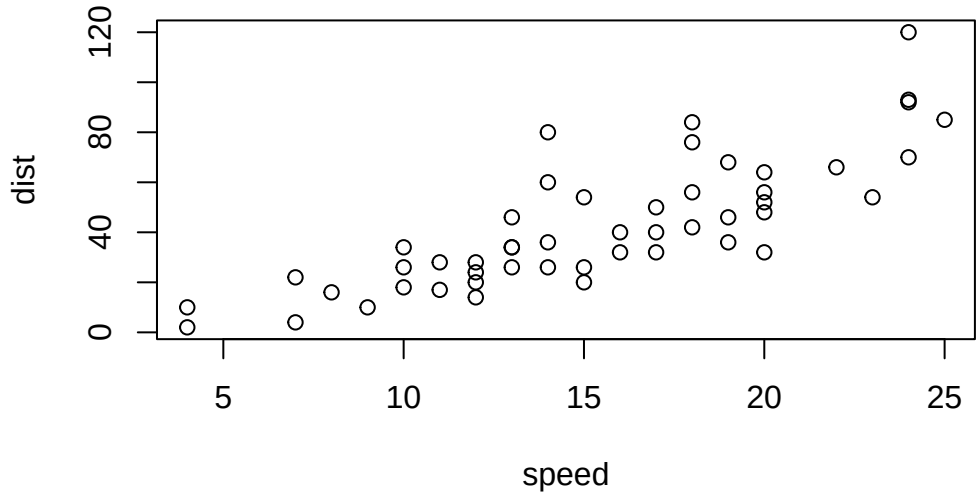
Here is a simple "base" R plot.

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

We can simply pass this to the `plot()` function.

```
plot(cars)
```



Key-point: Base R is quick but not so nice looking in some folks' eyes

Let's see how we can plot this with **ggplot**

1st I need to install this add-on package. For this we use the `install.packages()` function - *WE DO THIS IN THE CONSOLE, NOT our report.* This is a one time only deal.

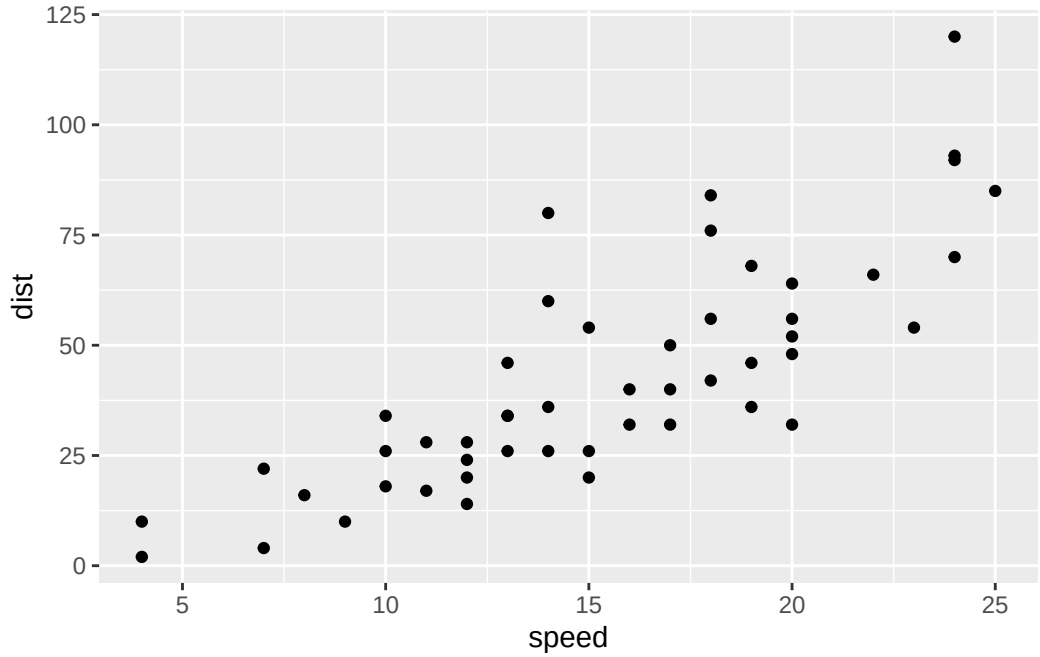
2nd We need to load the package with the `library()` function every time we want to use it.

```
library(ggplot2)
```

Every ggplot is composed of at least 3 layers:

- **data** (i.e. a data.frame with the things you want to plot)
- aesthetics **aes()** that map the columns of data to your plot features (i.e. aesthetics)
- geoms like **geom_point()** that srt how the plot appears.

```
ggplot(cars) +
  aes(x=speed, y=dist)+
  geom_point()
```



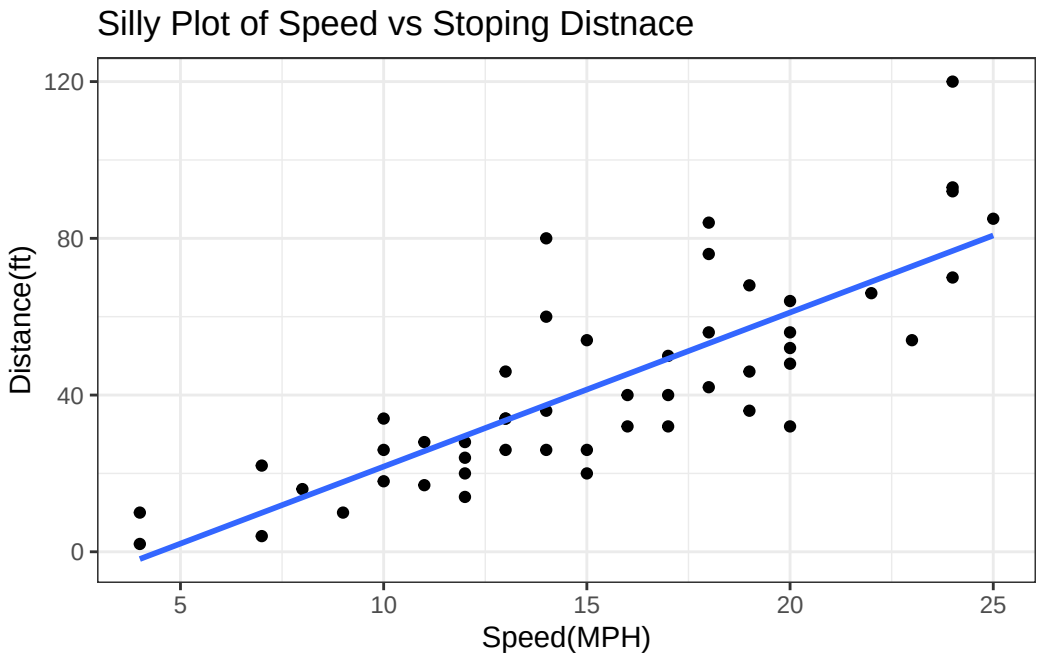
Key-point: For simple "canned" graphs base R is quicker but as things get more custom and elaborate then ggplot wins out…

Let's add more layers to out ggplot

Add a line showing the relationship between x and y Add title of the graph Add titles for x and y axis Change theme

```
ggplot(cars) +
  aes(x=speed, y=dist)+
  geom_point()+
  geom_smooth(method="lm", se=FALSE)+
  labs(title="Silly Plot of Speed vs Stoping Distnace")+
  xlab("Speed(MPH)")+
  ylab("Distance(ft)")+
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Going Further

Read gene expression data

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

Q1. How many genes are in this dataset?

```
nrow(genes)
```

```
[1] 5196
```

Q2. How many “up” regulated genes are there?

```
sum(genes$State=="up")
```

```
[1] 127
```

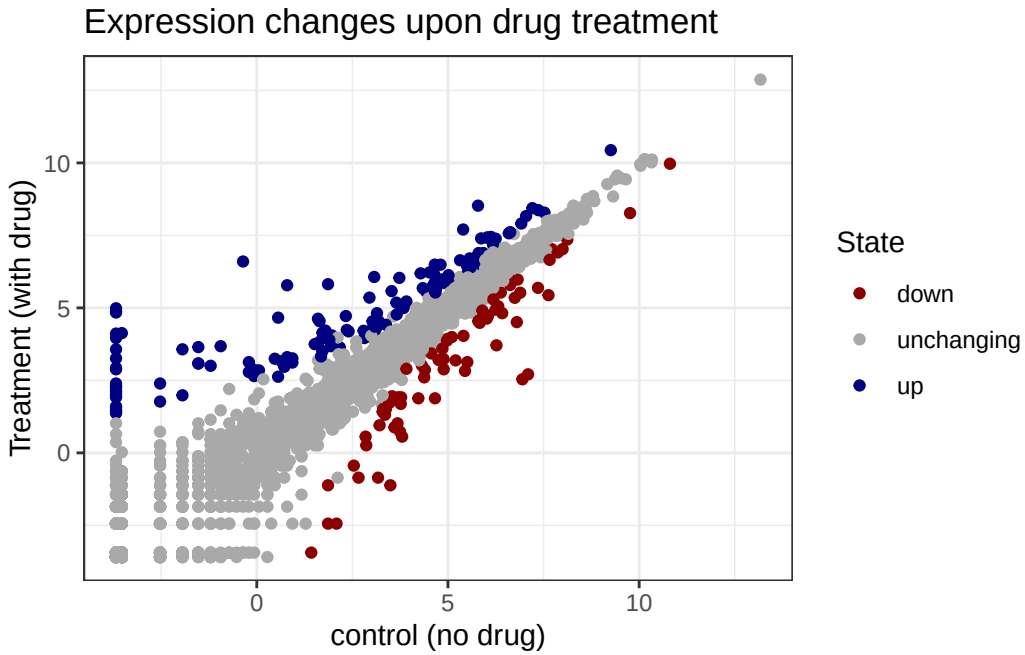
A useful function for counting up occurances of things in a vector is the `table()` function

```
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

```
p <- ggplot(genes)+
  aes(x=Condition1,y=Condition2)+
  geom_point()+
  aes(col=State)

p +
   scale_color_manual(values=c("darkred","darkgrey","navy"))+
  labs(title="Expression changes upon drug treatment",
       x="control (no drug)",
       y="Treatment (with drug)")+
  theme_bw()
```




```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
```

```
head(gapminder,3)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
```

```
tail(gapminder,3)
```

```
      country continent year lifeExp      pop gdpPercap
1702 Zimbabwe    Africa 1997  46.809 11404948  792.4500
1703 Zimbabwe    Africa 2002  39.989 11926563  672.0386
1704 Zimbabwe    Africa 2007  43.487 12311143  469.7093
```

Q4. How many countries are in this data set?

```
length(table(gapminder$country))
```

```
[1] 142
```

Q4. How many continents are in this data set?

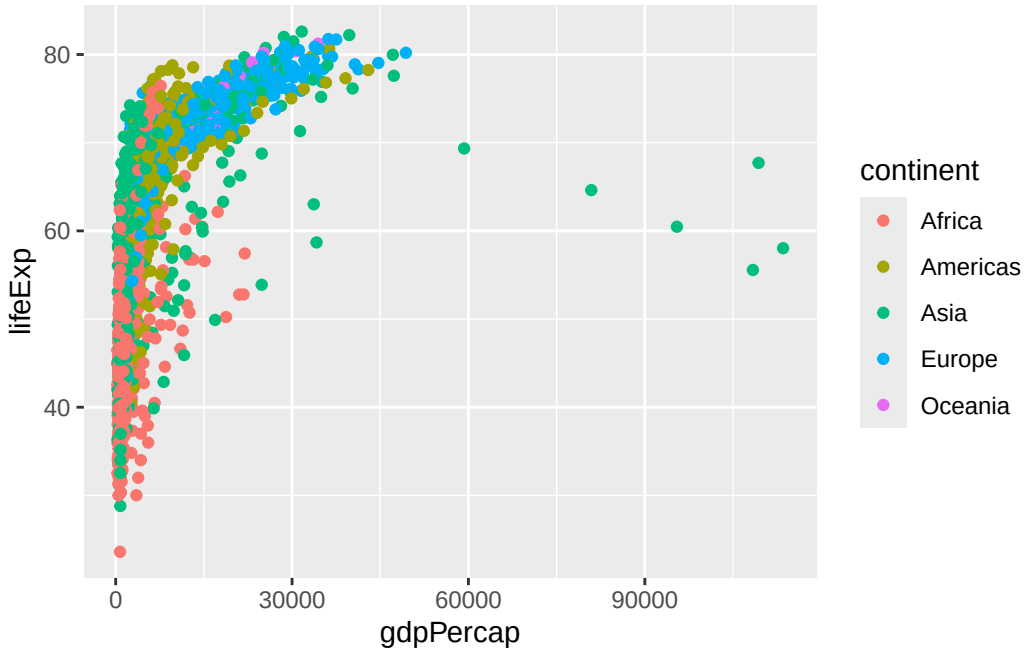
```
unique(gapminder$continent)
```

```
[1] "Asia"     "Europe"   "Africa"   "Americas" "Oceania"
```

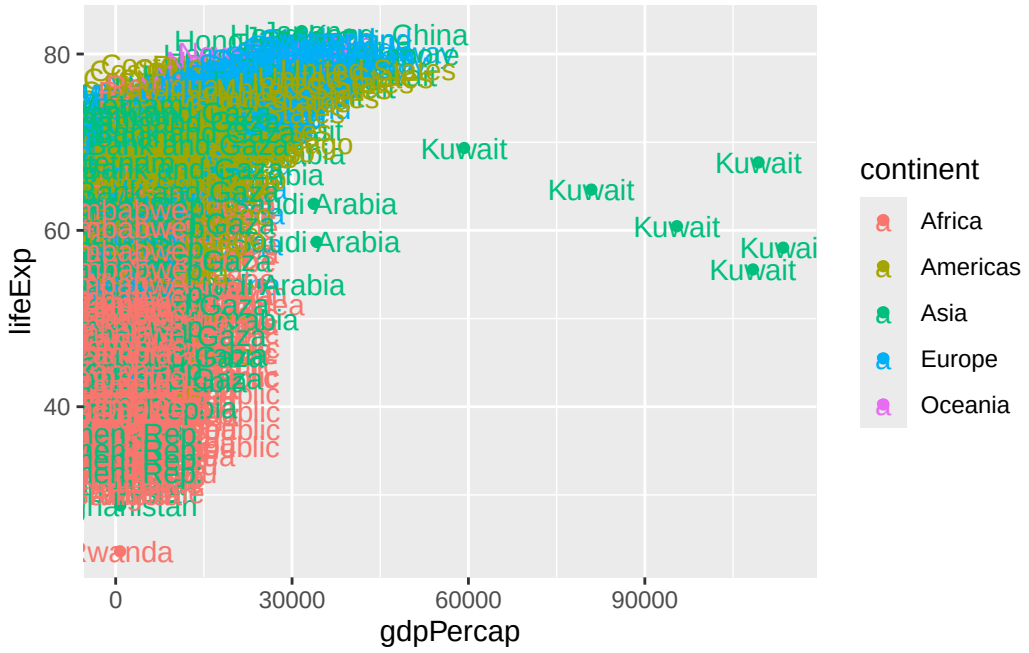
```
length(table(gapminder$continent))
```

```
[1] 5
```

```
ggplot(gapminder)+
  aes(x=gdpPercap, y=lifeExp,
      col=continent)+
  geom_point()
```



```
ggplot(gapminder)+
  aes(x=gdpPercap, y=lifeExp,
      col=continent,
      label=country)+
  geom_point()+
  geom_text()
```



I can you **ggrepel** package to make more sensible labels here.

```
library(ggrepel)
```

I want seperate pannel per continent

```
ggplot(gapminder)+
  aes(x=gdpPercap, y=lifeExp,
      col=continent,
      label=country)+
  geom_point()+
  geom_text_repel()+
  facet_wrap(~continent)
```

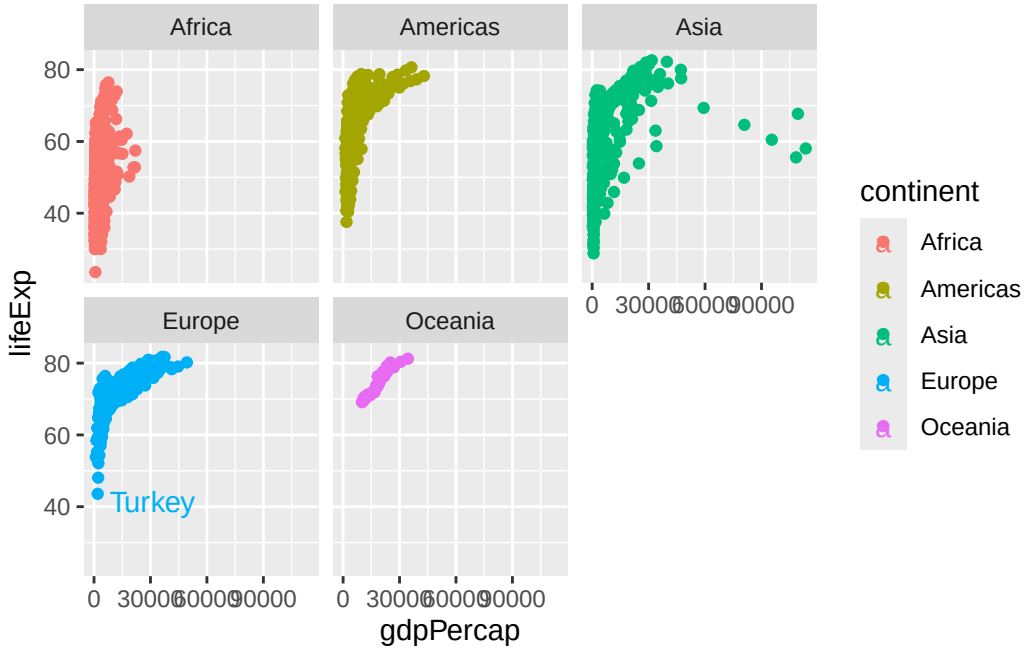
```
Warning: ggrepel: 624 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```

```
Warning: ggrepel: 359 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```

```
Warning: ggrepel: 300 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```

```
Warning: ggrepel: 24 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```

```
Warning: ggrepel: 396 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



Summary

ggplot2 makes it easier to create publication-quality, beautiful figures and allows you to build plots layer by layer using a consistent grammar. It is more concise for complex plots, offers sensible defaults, and handles legends and layout automatically. Base R plots are faster for simple, exploratory graphs but are harder to customize and refine for polished results. ggplot2 is preferred for finalized, professional visualizations, while base R is often used for quick, preliminary plots [1], [2], [3], [5], [4].